In-Context Learning with Differentially Private Text Sanitization in Large Language Models

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Abstract. With the increasing popularity of cloud-based large language models, users often inadvertently input text containing personal information during interactions, leading to significant privacy concerns. To address this challenge, we propose an in-context learning (ICL) based on differential privacy (DP) to protect users' instances and context information formally. The core idea entails enhancing local differentially private text sanitization and using token mapping relationships to remap private responses effectively. We conduct the experimental evaluation by comparing our method with Custext, two-shot, and zero-shot. The findings indicate that our method can attain a competitive edge while maintaining robust privacy protections. Code is available at https://github.com/larryfans/ICL_DP_Text-Sanitization.

Keywords: In-Context Learning · Differential Privacy · Large Language Model

1 Introduction

In recent years, there has been an explosive development in the large language model (LLM), with a series of large language models (LLMs) such as ChatGPT [1] and LLaMA [21] being introduced. LLMs have been applied to a wide range of language-related tasks, including question-answering [11], extracting core content [17], solving deliberate problems [25], etc.

Despite this advantage, a significant private challenge arises when users include personal or sensitive information within these prompts to pursue satisfactory responses. This practice introduces privacy risks, as the LLMs can extract and potentially expose embedded private knowledge [15]. Amazon serves as an example [10]. The company has warned its workforce, advising them against sharing proprietary data with ChatGPT. The cloud-based Large Language Models (LLMs) can easily extract private knowledge embedded within the prompts, thereby raising substantial privacy concerns.

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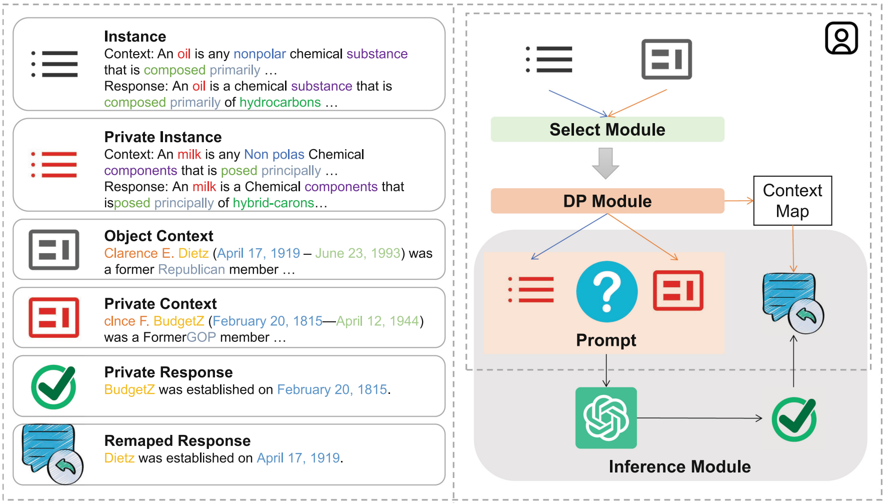

Fig. 1. The overview of our differential privacy ICL methodology. On the left side, we also use the same word color to highlight the changes in input text before and after perturbation and the changes in response before and after mapping.

In light of privacy issues, differential privacy (DP) [7], widely utilized across various privacy application domains [8,16,24], is acknowledged as the paramount standard for mitigating privacy-related challenges. Meanwhile, some works [6,9,19,20] have initiated a focus on the application of DP to safeguard private knowledge. A notable limitation of client-side differential privacy mechanisms is that noise injection often necessitates inference on LLMs, presenting a substantial challenge for users with limited computational capabilities.

Besides, DP text sanitization methods [3,26] can achieve noise injection using a word vector library(e.g., GloVe [14]) significantly smaller than the language model, enabling efficient implementation. Recent work by Tong et al. [20] introduced InferDPT, using DP text sanitization before querying cloud-based large language models. Nevertheless, their approach still necessitates using a local LLM to merge the original text with the responses from the cloud-based model to derive the final answer.

Despite advancements in differentially private text sanitization, we observed that existing DP text sanitization methods struggle to achieve the performance level of non-private scenarios when executing specific LLM tasks (e.g., summarization) under strong privacy guarantees.

To enhance the performance of LLMs on specific tasks, Brown et al. [2] proposed in-context learning (ICL) as a method to help LLMs adapt to domain-specific information. ICL entails adapting to downstream tasks without fintuning the weights of a pre-trained model [5].

To bridge the gap above, we introduce a novel framework that combines in-context learning with metric local differential private text sanitization to safeguard user privacy during interactions with cloud-based LLMs. To our knowledge, our work is the first to explore differential privacy text sanitization within in-context learning. Our method is presented in Fig. 1.

To address the limitations of existing methods that rely on local LLMs, we propose a token mapping mechanism to optimize generated responses, making privacy-preserving access to cloud-based large language models feasible for resource-constrained users.

Our experimental results demonstrate that our proposed method performs substantially better than existing approaches on specialized LLM tasks while simultaneously providing robust privacy protections.

2 Preliminary

Definition 1. Metric local differential privacy (Metric-LDP) [27]**:** An algorithm $\mathcal{M}(\cdot)$ adheres to (ε, d)-local differential privacy, for a given privacy parameter $\varepsilon \geq 0$, if for every pair of inputs $(t_1, t_2) \in T \times T$ satisfying a distance metric $D(t_1, t_2) \leq d$. Then, the probability Pr of any output $s \in S$,

$$Pr[\mathcal{M}(t_1) = s] \leq e^{\varepsilon \cdot d} Pr[\mathcal{M}(t_2) = s] \tag{1}$$

When $D(t_1, x_2) = 1$, $\forall x_1 \neq x_2$, Metric-LDP simplifies to LDP. Specifically, Metric-LDP will achieve indistinguishability between any two inputs within a metric distance less than d in the total metric space.

Due to the difficulty in defining absolute measurement distances in high-dimensional word vector spaces, the current practice [3,14] is to use the nearest neighbors metric, specifically the k-nearest tokens of a given token in the word vector space. In a word vector space of vocabulary number N, the distance between a token and its k-nearest neighbors can be expressed as $d = L/k$.

Definition 2. Exponential Mechanism [13]**:** A score function f: $T \times S \rightarrow \mathcal{R}$ be a randomized mechanism where for any inputs $t \in T$ and any possible outputs $s \in S$ we have:

$$Pr[s|t] \propto exp(\frac{\varepsilon \cdot f(t, s)}{2\triangle_f}), \tag{2}$$

where $\triangle_f$ is the sensitivity and $sup_{s \in S}|f(t, s) - f(t', s)| \leq \triangle_f$.

To ensure differential privacy, we apply the exponential mechanism to perturb the similarity scores of candidate tokens from S to S'. A token is then selected probabilistically based on these perturbed scores. To facilitate analysis, we normalize the score function f to a predetermined sensitivity $\triangle_f$(e.g., 1). Given a privacy parameter $\varepsilon \geq 0$, this process can be mathematically formalized as:

$$Pr[s|t] = \frac{exp(\frac{\varepsilon \cdot f(t,s)}{2\triangle_f}}{\sum_{s \in S'} exp(\frac{\varepsilon \cdot f(t,s)}{2\triangle_f}}). \tag{3}$$

3 Algorithmic Design

Threat Model. We assume that an adversary on the cloud-based model provider side can obtain the input raw text provided by the client. The adversary's goal is to extract sensitive information (e.g., personal privacy) from the input information on the client. In short, private identity information (e.g., names) can be exposed in the prompts. Empirical studies have confirmed the privacy risks of feasible attacks in existing work [6,23].

Our objective is to prevent adversaries from obtaining private information from user inputs while ensuring that users receive satisfactory answers. To mitigate the risk of exposing private information from user inputs while maintaining response quality, we introduce a novel private ICL framework that selectively perturbs only the privacy-sensitive components of examples and context based on a carefully designed prompt template. As illustrated in Fig. 1, our algorithm is deployed locally on the user's device and is divided into three primary modules. (1) select module, (2) DP module, and (3) inference module. It is worth mentioning that the instance and context will execute the select module and DP module, respectively, and finally aggregate into the input of the cloud LLM according to the prompt template in the inference module.

3.1 Select Module

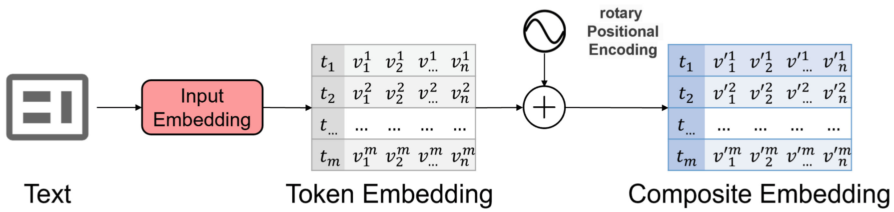

Fig. 2. The diagram illustrates the process of generating composite embedding by integrating token embedding with rotary positional encoding.

Words containing private information are typically found only among non-stop words, and not within stop words and symbols. The objective of the select module is to find the k-nearest tokens for each non-stop word in the text. To achieve this objective, it is necessary to utilize a token vector library $\mathcal{L} = \{t_i \mapsto (v_1^i, v_2^i, \ldots, v_n^i) \mid i = 1, 2, \ldots, l\}$, which encompasses l tokens and their corresponding n-dimensional vector representations. Notably, the token vector library $\mathcal{L}$ is derived from the collection of tokens within the tokenizer of the publicly accessible LLM and the corresponding input embedding layer.

First, we use the tokenizer integrated with LLMs to slice the user-side sentence into a tokens list $\mathcal{T} = [t_1, t_2, \ldots, t_m]$ containing m token numbers.

Subsequently, we transform the $\mathcal{T}$ into embedding vectors of shape $m \times n$ based on the input embedding layer. We observe that each token from $\mathcal{L}$ corresponds to only one vector. However, a single word can convey different semantics depending on its position within the text. For instance, the semantic meaning of "bank" differs significantly between "a large bank of knowledge" and "go to the bank". To enable language models to better understand textual semantics, [22] proposes the positional encoding technique to express the absolute position information of the token in the context. Consequently, We employ the more advanced rotary positional encoding method [18], in conjunction with the token embedding vectors, to obtain the composite embedding vectors $\mathcal{C} = \{t_i \mapsto (v_1'^i, v_2'^i, \ldots, v_n'^i) \mid i = 1, 2, \ldots, m\}$ that better capture the word semantics about the sentence context. The specific process is shown in Fig. 2.

Secondly, for each token in private content, we select the k most semantically relevant tokens from $\mathcal{L}$ based on $\mathcal{C}$. Semantic relevance can be determined by comparing the word vector similarities of two tokens, commonly using metrics such as Euclidean distance or cosine similarity. The k-nearest tokens $\{t_1', t_2', \ldots, t_k'\}$ and their corresponding similarity scores $\{s_1, s_2, \ldots, s_k\}$ for each token will be stored in a similarity token dictionary $S : \{t_1', t_2', \ldots, t_k'\} \rightarrow \{s_1, s_2, \ldots, s_k\}$ for subsequent use by the DP component.

3.2 DP Module

In the DP module, inspired by [3, 26], for each token in the user-side text $\mathcal{T}$ that contains sensitive information, we randomly select a candidate token t^* from the set S using the exponential mechanism 2. After undergoing metric-LDP 1 perturbation, the raw text $\mathcal{T}$ is transformed into a private text $\mathcal{T}'$.

Due to the substantial semantic gaps between noised text and original text caused by token replacements, existing methods that rely on LLMs to generate high-quality answers often face computational challenges, limiting their deployability on resource-constrained devices like mobile phones.

Storing the mapping between original and replaced tokens is a lightweight approach that can be effectively used to improve the efficiency of our algorithm without requiring significant computational resources. Specifically, for each interaction with the cloud-based LLMs, a mapping dictionary $D : \{t_1, t_2, \ldots, t_{m'}\} \rightarrow \{t_1^*, t_2^*, \ldots, t_{m'}^*\}$ is created to store the mapping between original tokens and their corresponding replaced tokens (where m' represents the number of unique tokens in an input sentence of length m). To avoid redundancy, only the first occurrence of a token is mapped.

3.3 Inference Module

We design a prompt template framework for ICL, as shown in Table 1. The framework consists of three components: a question, a context, and two instances. Users provide information for each component when interacting with the cloud-based LLM. The question specifies the exact task (e.g., summarization), the context provides direct content for the task, and the instances, composed of a

Table 1. The prompt template for In-Context Learning.

Context 1: {Context Instance 1}
Response 1: {Response Instance 1}
Context 2: {Context Instance 2}
Response 2: {Response Instance 2}
Question: The above are instances of summary sentences, please help me summarise the following sentence.
Context: {Context}
Response:

context and a reference answer, guide the LLM's response generation. Notably, the question section, highlighted in blue in the table, does not contain sensitive information and thus does not require DP perturbation. Conversely, highlighted in red, the context and instance sections often contain sensitive information and are subject to DP perturbation.

In the inference module, the instruction template guides the merging of the private instances $\mathcal{T'}_1, \mathcal{T'}_2$ and private context $\mathcal{T'}_3$ to form a single prompt. Next, the prompt is sent to a cloud-based LLM service provider (e.g., ChatGPT) and receives the model's response. Since the example and context texts have been sanitized, most entities, except for stop words, have been altered. Directly using the response as-is often fails to meet utility requirements. Fortunately, we have preserved the mapping relationships of the context, allowing us to revert the sanitized entities to their original form easily.

4 Evaluation

4.1 Experimental Setup

Datasets. We focused on the text summarization task and conducted experiments using 1,188 summarization category records from the databricks-dolly-15k [4] dataset. The databricks-dolly-15k dataset, with over 15,000 records, spans eight directive categories and showcases the interactivity of large language models. Furthermore, we split the summarization category records into an instance set and a test set in a 2:1 ratio.

Baseline. We juxtaposed our approach with CusText [3], a cutting-edge method known for text sanitization of DP. CusText used the Glove.42B.300d embedding model with 1917495 vocabulary and 300-dimensional vectors per token. Both mechanisms were compared under the same conditions of candidate token scope K and privacy budget ε for a fair assessment.

Implementation. Our mechanism used the Meta-Llama-3-8B-Instruct token library and word embedding layer with 128256 tokens and 4096-dimensional

Table 2. ROUGE results on different mechanisms, bold numbers indicate performance surpassing that of zero-shot prediction.

Metric		ROUGE1	ROUGE2	ROUGEL	ROUGELSum
Zero-shot	$\varepsilon = \infty$	0.375	0.176	0.282	0.293
Two-shot	$\varepsilon = \infty$	0.570	0.383	0.501	0.509
CusText	$\varepsilon = 1$	0.316	0.123	0.233	0.245
	$\varepsilon = 2$	0.320	0.124	0.236	0.248
	$\varepsilon = 3$	0.322	0.126	0.239	0.250
Ours(no map)	$\varepsilon = 1$	**0.382**	0.16	**0.336**	**0.341**
	$\varepsilon = 2$	**0.423**	**0.205**	**0.377**	**0.382**
	$\varepsilon = 3$	**0.514**	**0.309**	**0.461**	**0.466**
Ours(map)	$\varepsilon = 1$	**0.520**	**0.309**	**0.465**	**0.471**
	$\varepsilon = 2$	**0.534**	**0.331**	**0.481**	**0.486**
	$\varepsilon = 3$	**0.559**	**0.365**	**0.503**	**0.508**

vectors per token. For the cloud large language model, we use GPT3.5-turbo with the max new token set to 60 and the temperature parameter set to 0.1. Following [3], the metric for selecting the most semantically relevant candidates set cosine similarity and set k-nearest tokens to 20.

Indicator. To conduct a thorough evaluation of our methodologies, we employed a combination of ROUGE [12] and BERTScore [28] metrics. Elevated scores in these metrics are indicative of enhanced performance.

4.2 Application Utility

Text summarization is considered one of the most significant undertakings within the domain of LLMs. We aimed to evaluate the performance of text sanitization algorithms based on differential privacy in the context of text summarization tasks within the in-context learning scenario.

In the two-shot prediction, the procedure is implemented such that every two instances correspond to the context of one test set. For the two-shot prediction, we used the prompt from Table 1. For the zero-shot prediction, the instance portion was removed from Table 1, and the question was replaced with: “Please help me summarize the following sentence”.

Within the two-shot prediction, we compared our methodology with the non-DP zero-shot prediction $\varepsilon = \infty$, non-DP two-shot prediction $\varepsilon = \infty$, and the CusTxt mechanism. Furthermore, we juxtaposed the performance of our methodology in responses devoid of context maps with those that incorporate context maps.

Results. We present the main results in Table 2 and Table 3 for privacy levels $\varepsilon = \{1, 2, 3\}$. The CusText mechanism does not achieve the effectiveness

Table 3. BERTScore results on different mechanisms, bold numbers indicate performance surpassing that of zero-shot prediction.

Metric		Precision	Recall	F1
Zero-shot	$\varepsilon = \infty$	0.825	0.807	0.814
Two-shot	$\varepsilon = \infty$	0.920	0.864	0.890
CusText	$\varepsilon = 1$	0.805	0.791	0.796
	$\varepsilon = 2$	0.805	0.791	0.796
	$\varepsilon = 3$	0.807	0.794	0.799
Ours(no map)	$\varepsilon = 1$	**0.850**	**0.814**	**0.831**
	$\varepsilon = 2$	**0.866**	**0.826**	**0.845**
	$\varepsilon = 3$	**0.903**	**0.853**	**0.876**
Ours(map)	$\varepsilon = 1$	**0.895**	**0.851**	**0.872**
	$\varepsilon = 2$	**0.902**	**0.854**	**0.877**
	$\varepsilon = 3$	**0.915**	**0.862**	**0.887**

of non-DP zero-shot prediction. However, our methodology consistently outperforms the non-DP zero-shot prediction in nearly all instances. Additionally, the performance of our methodology, after incorporating context maps, significantly surpasses that without context maps. In fact, under the setting of $\varepsilon = 3$, the performance is extremely close to two-shot prediction.

5 Conclusion

Existing methods struggle to simultaneously satisfy the requirements of user privacy protection, low-computation environments, and practicality when using cloud LLMs. Our framework incorporates in-context learning to provide privacy examples. It selectively applies local differential privacy text sanitization to private contexts and examples, further enhanced by a novel word mapping technique that significantly improves the quality of responses from cloud-based LLMs. The experiment proved our method maintains robust privacy protections and achieves a competitive edge in performance. Integrating differential privacy with in-context learning, as proposed in this paper, represents a significant advancement in the field.

Limitations and Future Work. One limitation of our current work is the sequential arrangement of privacy examples in the training set. Future research could explore retrieval-augmented generation to identify optimal examples from a knowledge base, potentially improving response quality. Additionally, our study focuses on word-level differential privacy, which offers a weaker privacy guarantee than sentence-level DP. Investigating sentence-level DP for text sanitization is a promising avenue for future research. Furthermore, our current work is confined to the summarization task for LLMs and we plan to apply our method in future studies on a broader range of ICL tasks.

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